

WEEK8

Aim:

To develop a simple **Mini Web Application using Node.js and Express.js** with registration, login, dashboard and JSON file storage.

Description:

Develop a **Mini Web Application using Node.js and Express.js** based on a real-world application scenario. The application consists of a **Home/Landing Page, Registration Page, Login Page, and User Dashboard**. Users can register by entering their details, and the registration data is sent to the Express.js backend using a **POST request** and stored in a **JSON file** as a simple database. During login, the entered username and password are checked against the stored user details. If the credentials are valid, the user is allowed to access the dashboard; otherwise, an error message is displayed. The application also provides a **logout option** to return to the login page.

Program Code:

1) app.js:

```
const express = require("express");
const fs = require("fs");
const path = require("path");

const app = express();
const PORT = 3000;

app.use(express.urlencoded({ extended: true }));
app.use(express.json());

app.use(express.static(path.join(__dirname, "public")));

// Home
app.get("/", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "index.html"));
});

// Register page
app.get("/register", (req, res) => {
  res.sendFile(path.join(__dirname, "public", "register.html"));
});

// Register user
```

```

app.post("/register", (req, res) => {

  const newUser = {
    name: req.body.name,
    age: req.body.age,
    dob: req.body.dob,
    gender: req.body.gender,
    email: req.body.email,
    mobile: req.body.mobile,
    username: req.body.username,
    password: req.body.password,
    address: req.body.address
  };

  fs.readFile("users.json", "utf8", (err, data) => { let users = [];

    if (!err && data.trim() !== "") {
      users = JSON.parse(data);
    }

    users.push(newUser);

    fs.writeFile(
      "users.json",
      JSON.stringify(users, null, 2),
      (err) => {
        if (err) {
          res.send("Error saving data");
        } else {
          res.send(
            <h2>Registration Successful!</h2> <a href="/login">Go to
            Login</a> `);
        }
      }
    );
  });

  // Login page
  app.get("/login", (req, res) => {
    res.sendFile(path.join(__dirname, "public", "login.html")); });

  // Login
  app.post("/login", (req, res) => {

```

```

const username = req.body.username;
const password = req.body.password;

fs.readFile("users.json", "utf8", (err, data) => {

  if (err) {
    res.send("Error reading data");
    return;
  }

  const users = JSON.parse(data);

  const user = users.find(
    u => u.username === username &&
    u.password === password
  );
  if (user) {
    res.sendFile(
      path.join(__dirname, "public", "dashboard.html") );
  } else {
    res.send(`
    <h2>Invalid Username or Password</h2>
    <a href="/login">Try Again</a>
    `);
  }
});

// Logout
app.get("/logout", (req, res) => {
  res.redirect("/login");
});

// Start server
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`); });

```

2)index.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Mini Web Application</title> <link
```

```
rel="stylesheet" href="style.css"> </head>
```

```
<body>
```

```
  <header>
```

```
    <h1>Mini Web Application</h1>
```

```
    <nav>
```

```
      <a href="/">Home</a>
```

```
      <a href="/register">Register</a>
```

```
      <a href="/login">Login</a>
```

```
    </nav>
```

```
  </header>
```

```
<main class="container">
```

```
  <h2>Welcome to Our Application</h2> <p>
```

```
    Welcome to our Mini Web Application. </p>
```

```
  <p>
```

```
    This application is developed using
```

HTML, CSS, JavaScript, Node.js and Express.js. </p>

<p>

New users can register and existing users

can login to access the dashboard.

</p>

<div class="buttons">

Register Now Login

</div>

</main>

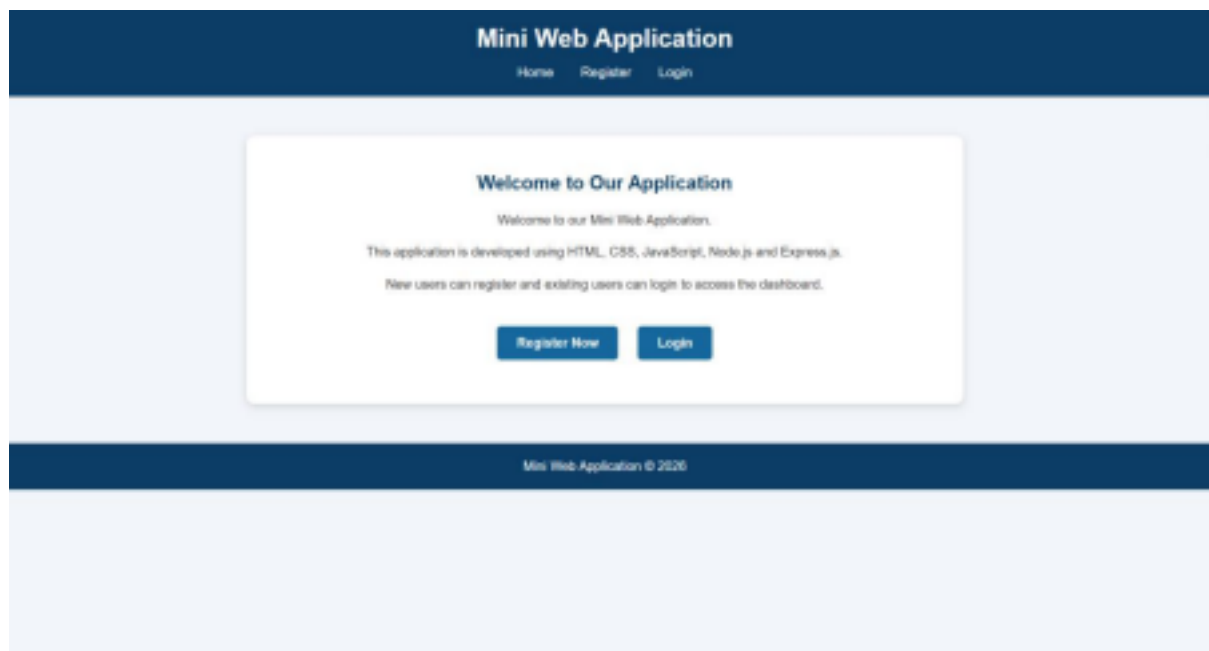
<footer>

<p>Mini Web Application © 2026</p> </footer>

</body>

</html>

Output:



3)register.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">
```

```
  <title>User Registration</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <header>
```

```
    <h1>User Registration</h1>
```

```
    <nav>
```

```
      <a href="/">Home</a>
```

```
<a href="/register">Register</a>
```

```
<a href="/login">Login</a>
```

```
</nav>
```

```
</header>
```

```
<main class="form-container">
```

```
<h2>Create Your Account</h2>
```

```
<form action="/register" method="POST">
```

```
<label>Name:</label>
```

```
<input
  type="text"
  name="name"
  placeholder="Enter your name"
  required
>
```

```
<label>Age:</label>
```

```
<input
  type="number"
  name="age"
  placeholder="Enter your age"
  required
>
```

```
<label>Date of Birth:</label>
```

```
<input
  type="date"
  name="dob"
  required
>
```

```
<label>Gender:</label>
```

```
<select name="gender" required>
```

```
<option value="">
```

Select Gender

</option>

<option value="Male">

Male

</option>

<option value="Female">

Female

</option>

<option value="Other">

Other

</option>

</select>

<label>Email:</label>

<input

type="email"

name="email"

placeholder="Enter your email"

required

>

<label>Mobile Number:</label>

<input

type="tel"

name="mobile"

placeholder="Enter mobile number" required

>

<label>Username:</label>

<input

type="text"

name="username"

placeholder="Create username"

required

>

<label>Password:</label>

<input

```
        type="password"
        name="password"
        placeholder="Create password"
        required
    >
```

```
<label>Address:</label>
```

```
<textarea
    name="address"
    rows="4"
    placeholder="Enter your address"
    required></textarea>
```

```
<button type="submit">
    Register
</button>
```

```
</form>
<p class="bottom-text">
```

Already have an account?

```
<a href="/login">
    Login here
</a>
```

```
</p>
```

```
</main>
```

```
</body>
```

```
</html>
```

Output:

User Registration

[Home](#)[Register](#)[Login](#)

Create Your Account

Name:

Enter your name

Age:

Enter your age

Date of Birth:

dd-mm-yyyy

Gender:

Select Gender

Email:

Enter your email

Mobile Number:

Enter mobile number

Username:

Create username

Password:

Enter your age

Date of Birth:

dd-mm-yyyy

Gender:

Select Gender

Email:

Enter your email

Mobile Number:

Enter mobile number

Username:

Create username

Password:

Create password

Address:

Enter your address

Register

Already have an account? [Login here](#)

4)login.html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Login</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>

<header>
<h1>User Login</h1>

<nav>
<a href="/">Home</a>
<a href="/register">Register</a>
<a href="/login">Login</a>
</nav>
</header>

<div class="form-container">
<h2>Login</h2>

<form action="/login" method="POST">

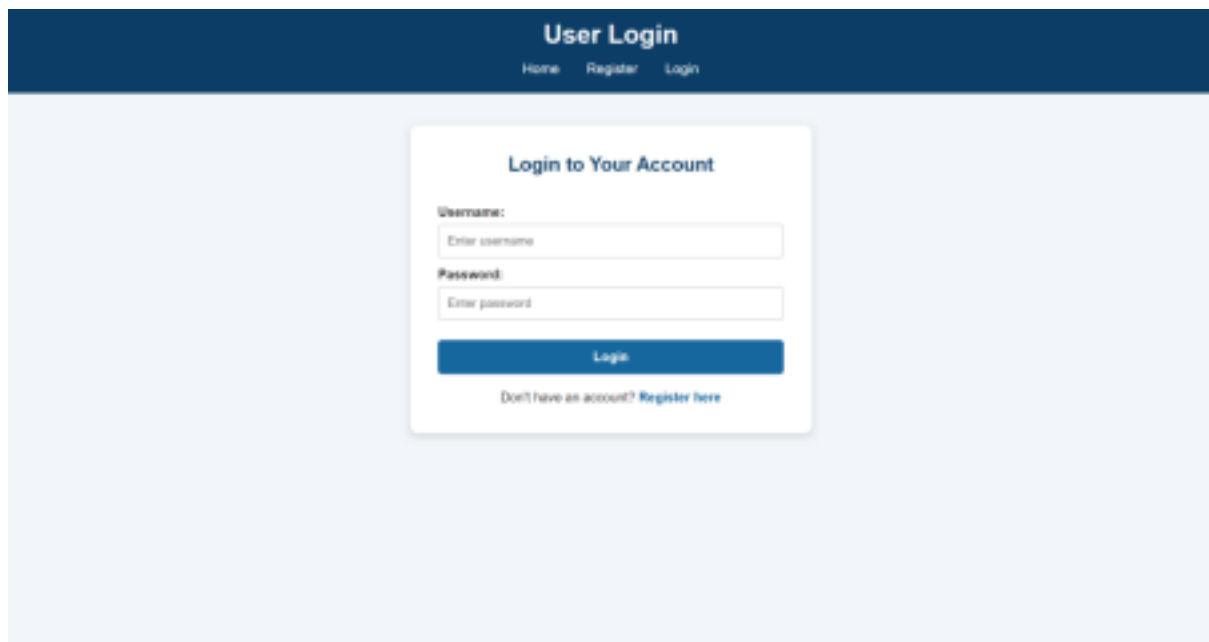
<label>Username:</label>
<input type="text" name="username" required>

<label>Password:</label>
<input type="password" name="password" required>

<button type="submit">Login</button>

</form>
<p>
Don't have an account?
<a href="/register">Register</a>
</p>
</div>
</body>
</html>
```

Output:



Style.css:

```
* {  
  box-sizing: border-box; }
```

```
body {  
  font-family: Arial;  
  margin: 0;  
  background-color: #f2f6fa; }
```

```
header {  
  background-color: #0c3d66; color:  
  white;  
  text-align: center;  
  padding: 20px;  
}
```

```
nav a {  
  color: white;  
  text-decoration: none;  
  margin: 15px;  
}
```

```
.container {  
  width: 80%;  
  margin: 50px auto;
```

```
padding: 40px;
background-color: white;
text-align: center;
border-radius: 10px;
}
```

```
.form-container {
width: 500px;
max-width: 90%;
margin: 40px auto;
padding: 30px;
background-color: white;
border-radius: 10px;
}
```

```
form {
display: flex;
flex-direction: column; }
```

```
label {
margin-top: 10px;
margin-bottom: 5px;
font-weight: bold;
}
```

```
input,
select,
textarea {
padding: 10px;
border: 1px solid #ccc;
border-radius: 5px;
}
```

```
button,
.btn {
margin-top: 20px;
padding: 12px;
background-color: #15679d; color:
white;
border: none;
border-radius: 5px;
text-decoration: none; cursor:
pointer;
}
```

```
}  
button:hover,  
.btn:hover {  
background-color: #0c3d66;  
}
```

Users.json:

```
[]
```

Result: Thus, the Mini Web Application using Node.js and Express.js was successfully developed and executed. The application allows users to register, store their details in a JSON file, login using valid credentials, access the dashboard, and logout.